

Container Terminal Resource and Performance Comparison of Self-Propelled Autonomous Railcars and Conventional Intermodal Trains

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BACKGROUND AND RESEARCH QUESTIONS

Self-Propelled Autonomous Railcars (SPARCs) vs. Conventional Trains (CTs) in Intermodal Service

	SPARCs	CTs
Size	20 to 50 vehicles	100 to 200 railcars
Properties	More frequent departure Less shipment waiting time Flexible platooning	Suited for high-volume long-haul corridors Long shipment dwell time Rigid train formation

Research Questions: Container Terminals for SPARCs vs. CTs

- How much intermodal terminal land area footprint can be eliminated?
- What reductions in terminal processing time and energy consumption can be achieved?
- How do the required terminal equipment resource levels compare to achieve equivalent performance?

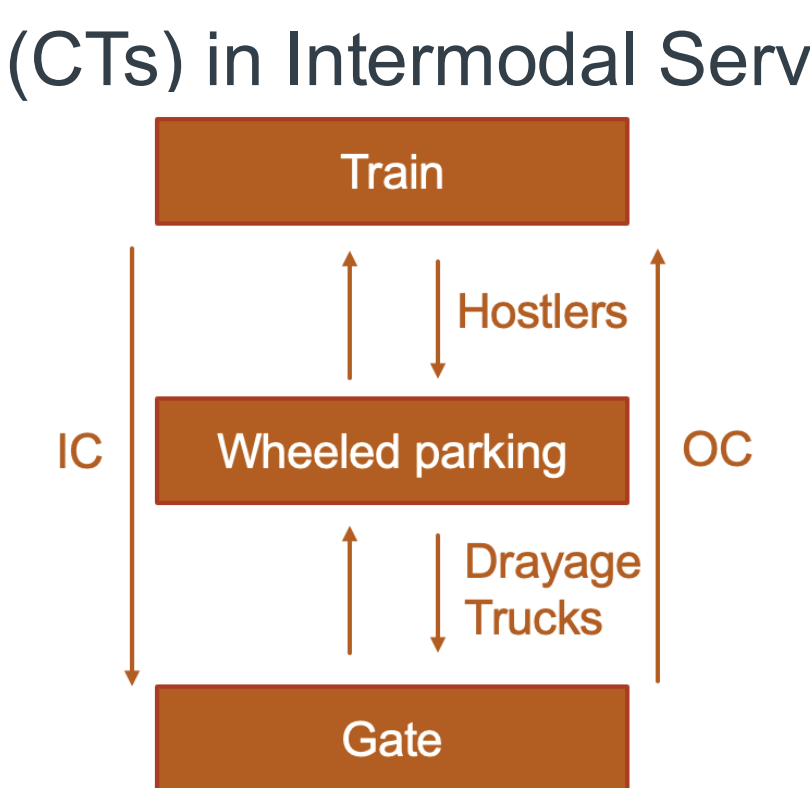


Fig. 1: Container terminal operation

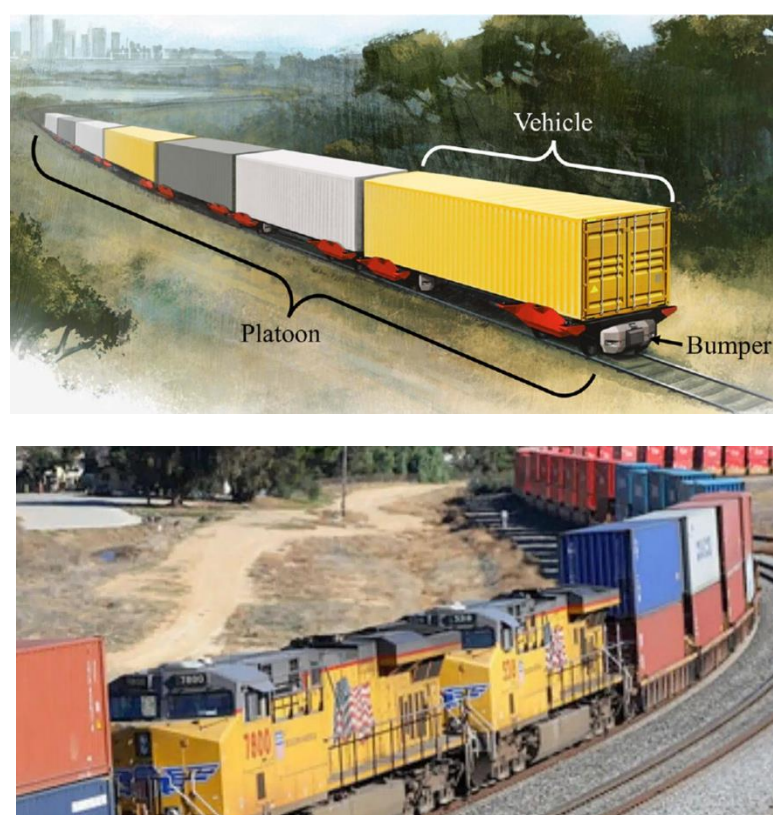


Fig. 2: SPARC Platoon (top) and Conventional Intermodal Train (bottom)

METHODOLOGY

Rail-Truck Terminal Working Cycle and Processes

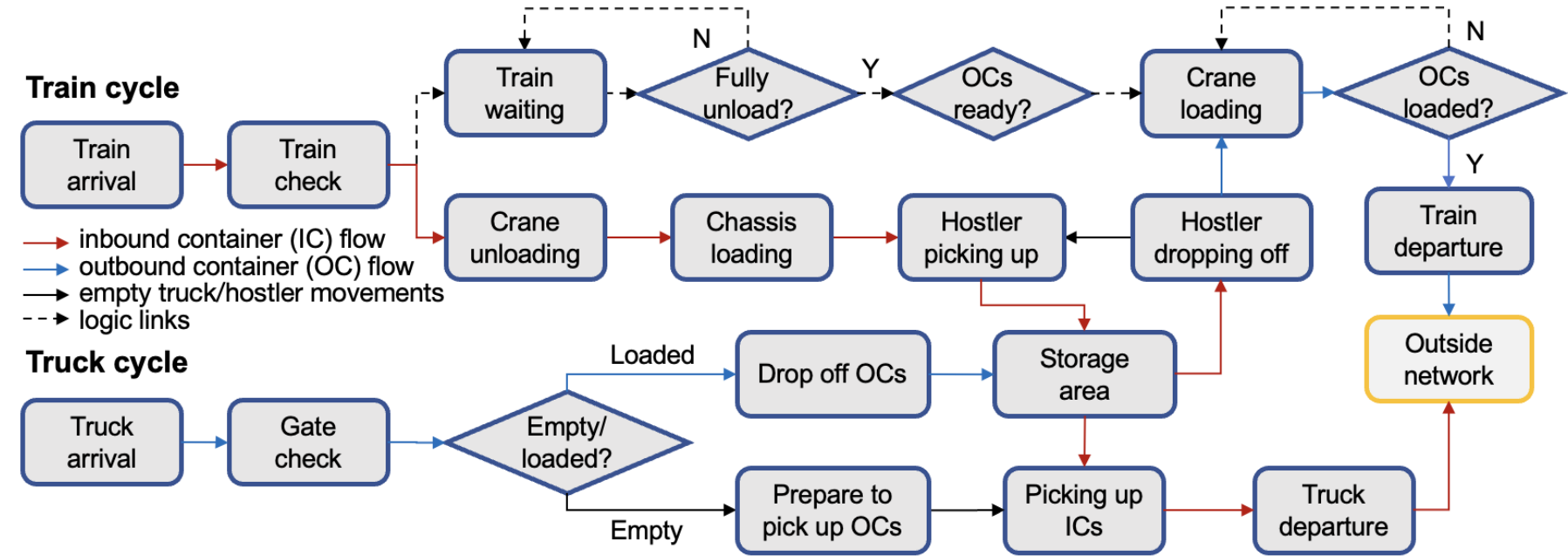


Fig. 3: High-level flow of processes considered in container terminal analysis

Optimal Container Terminal Layout Model

SPARCs and CTs can use different intermodal terminals optimized to the number of containers transported by each platoon/train.

Optimal terminal layout model assumptions:

- Terminal footprint is a rectangular area.
- Parking blocks parallel to single transload track.
- Evenly distributed parking blocks and aisle groups.
- Train length is less than the overall terminal length.
- Each parking block is identical.
- Gate is located at lower left corner in parking area.

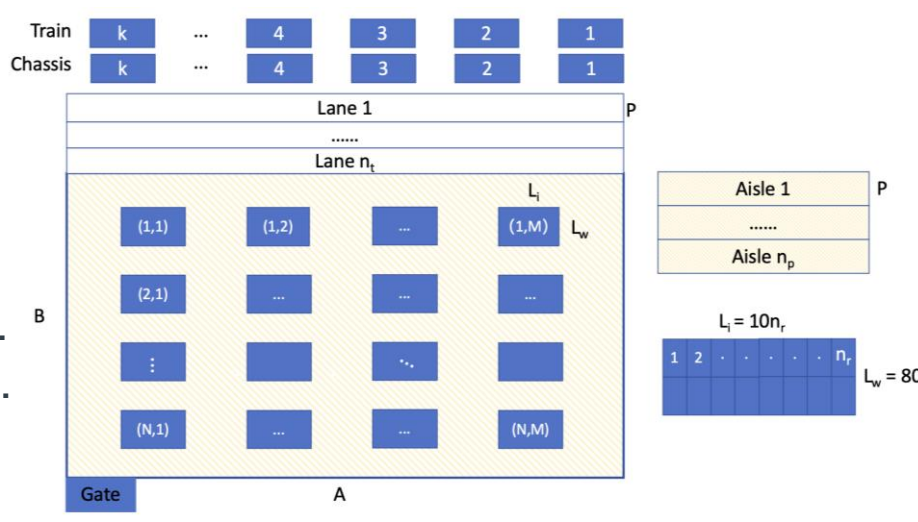


Fig. 4: Single-stack optimal layout

Optimization model minimizes the sum of three different terminal travel distances: hostler total travel distance (d_h), drayage truck travel distance (d_t), the total repositioning distance (d_r).

$$\text{Min } d(M, N, n_p, n_r) = d_h(M, N, n_p, n_r) + d_r(M, N, n_p, n_r) + d_t(M, N, n_p, n_r) \quad (1)$$

s.t.

$$2c_p M N n_r \geq k \quad (2)$$

$$0 \leq (M+1)n_p P + 10M n_r \leq A \quad (3)$$

$$0 \leq (N+1)n_p P + N L_w \leq B \quad (4)$$

$$M, N, n_p, n_r, n_r \in \mathbb{N}^* \quad (5)$$

Model decision variables:

- M : columns of parking blocks
- N : rows of parking blocks
- n_r : aisles per group on the train side
- n_p : aisles per group in the parking area
- n_r : pairs of "back-to-back" parking spots within a parking block

User-input model parameters:

- k : expected peak container storage demand
- A : maximum terminal length (in feet) parallel to the transload track
- B : maximum terminal width (in feet) perpendicular to the transload track

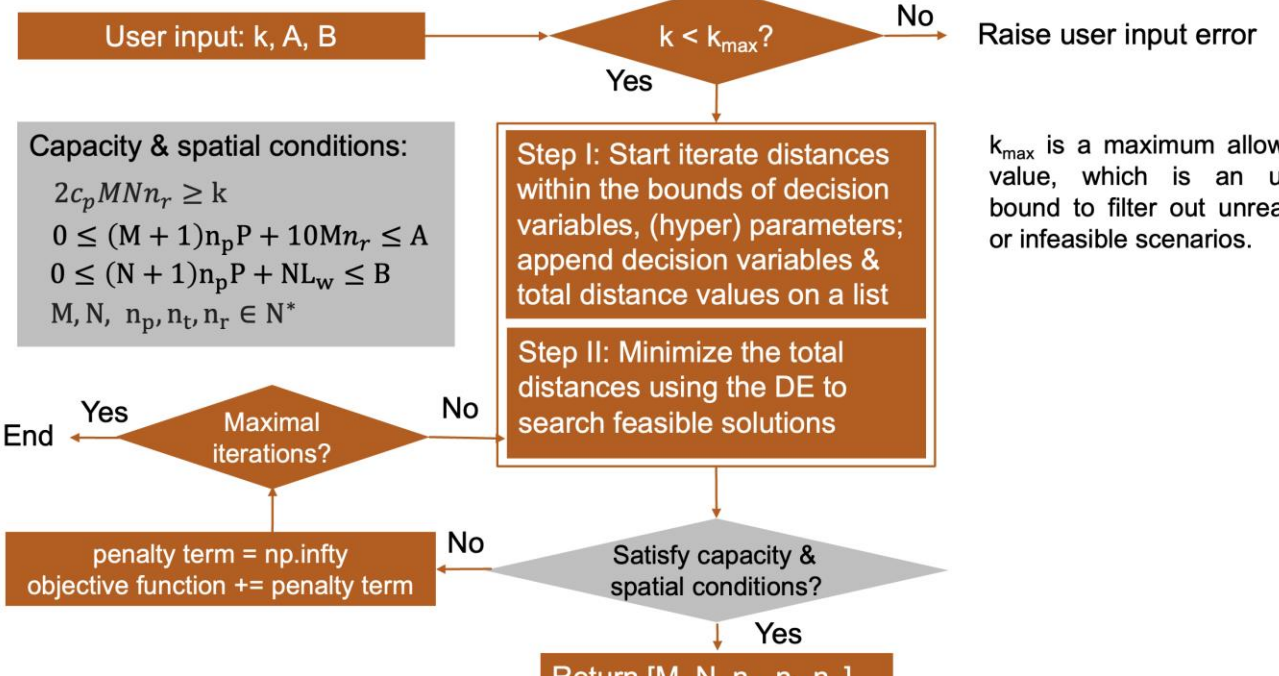


Fig. 5: Differential Evolution (DE) solution approach for container layout non-linear mixed integer optimization model

LIFTS Modeling Framework

Line-Haul Intermodal Freight Terminal Simulation (LIFTS) model developed in SimPy is used to evaluate and compare the performance of operations with SPARCs and CTs with their respective optimal container terminal layouts and train/platoon batch size

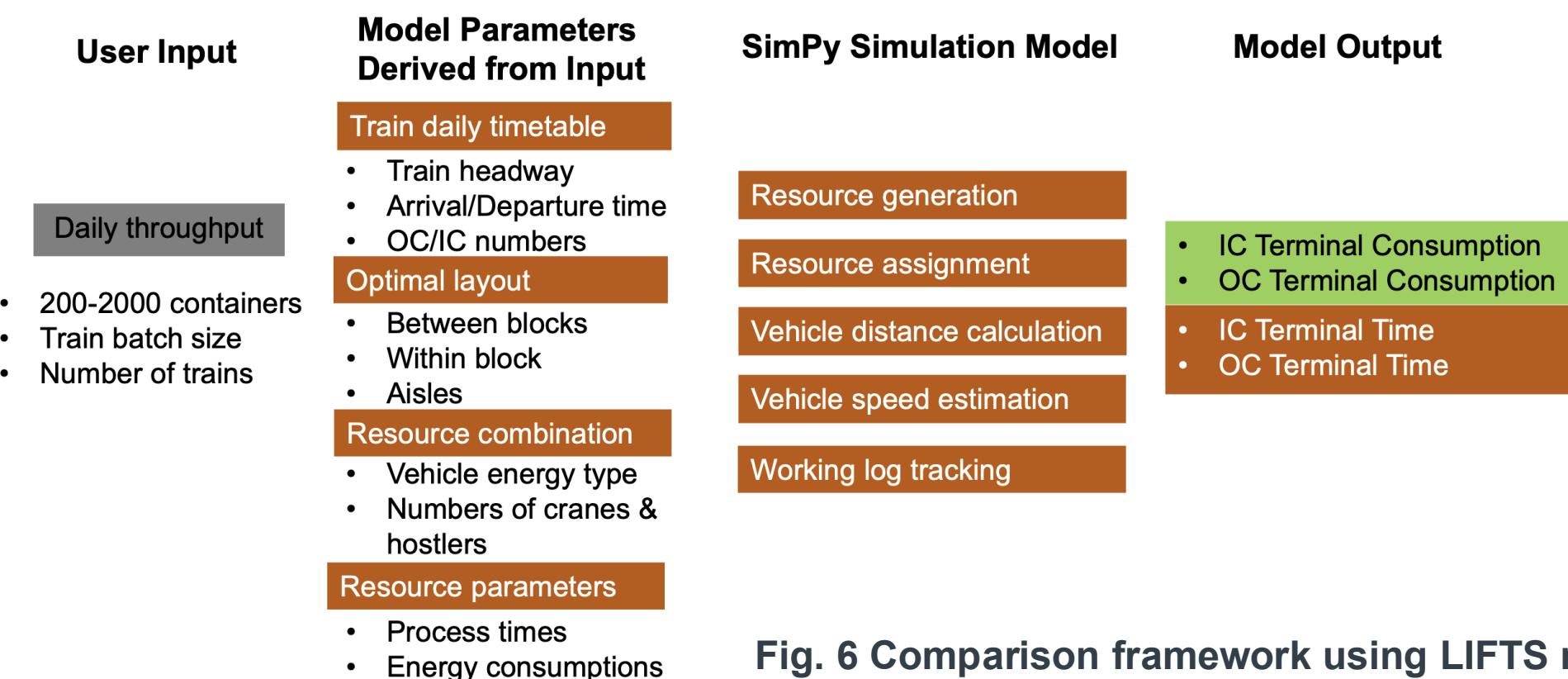


Fig. 6: Comparison framework using LIFTS model

Vehicle Speed-Density Estimation

LIFTS model considers the impact of terminal congestion on hostler and truck travel speeds and the time required to complete each step of the terminal process. Functions relating vehicle speed to traffic density were derived from more detailed AnyLogic container terminal model that considers vehicle routing conflicts.

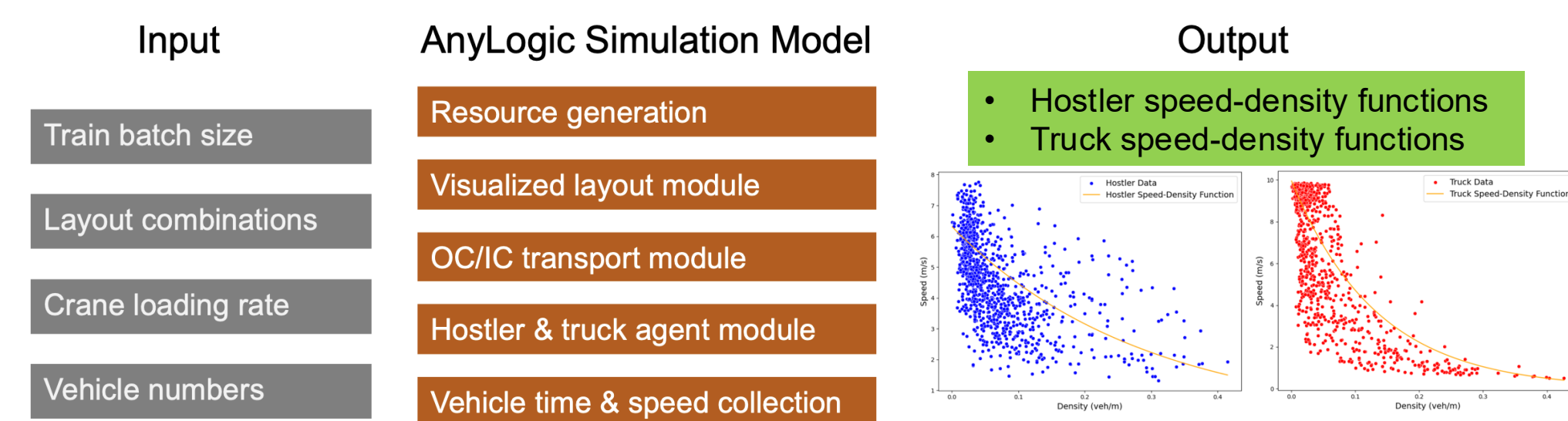


Fig. 7: Vehicle speed estimation framework

LIFTS Output

Average total terminal time (t) is defined as the sum of two key components: average processing time (t_1) and average delay time (t_2), $t = t_1 + t_2$. The t_1 term reflects the congestion and efficiency of terminal resources, and The t_2 term captures congestion and scheduling inefficiencies

$$t_1 = a_1 + b_1 \times \text{train batch size}$$

$$t_2 = a_2 e^{b_2} \times \text{number of trains}$$

Energy Estimation

To estimate the total energy consumption across the intermodal terminal, an energy-based accounting method is adopted using energy consumed per unit lift/load for diesel vehicles.

Table 1 Energy Consumption Matrices

Crane	Idle (load)	Loaded (load)
Diesel	0.70	4.10
Hostler	Idle (trip)	Loaded (trip)
Diesel	1.11	1.94
Truck	Idle (trip)	Loaded (trip)
Diesel	1.11	1.94

Note: Diesel unit is gallons/hour

EXPERIMENTAL DESIGN

LIFTS Input Data and Simulation Parameters

Container Volume and Traffic Demand

- Train batch size (k):** inbound containers (IC) per train or platoon
SPARCs: 20 to 50; CTs: 100 or 200; (assume one container per vehicle or railcar)
- Trains or Platoons per day:** 1, 2, 3, 4, 5, etc.
- Daily throughput (K):** $2 \times \text{trains/day} \times \text{containers/train}$; (IC + Outbound Containers)
- Terminal Layout Parameters**
 - Calculated from optimal layout module given train/platoon batch size (k)
- Terminal Resource Parameters**
 - Cranes: 1 (baseline), range from 1 to 4
 - Hostlers: 8 (baseline), range from 1 to 32
 - Trucks: Depends on the maximum of IC and OC, which remains relatively fixed.

Train/Platoon Timetable

- Randomize arrival times each day according to traffic level
- Arrival time: Noon +/- (12 hours $\times$ random uniform distribution between 0 and 1)
- Departure time: Equal to next train arrival time

Simulation Duration: 7 days minimum; extended if not fully processed (e.g. 10 days)

Data Collection Period: 3 days (the 2nd – 4th day of the simulation)

LIFTS Output Data

Inbound Containers (IC)

IC avg delay time: actual train arrival time (simulation log time) - expected train arrival time (train timetable)

IC avg processing time (no delay): truck exit time - train arrival time for each IC

IC avg terminal time: IC avg delay time + IC avg processing time (no delay)

IC emissions: weighted vehicle lifts + trips

Outbound Containers (OC)

OC avg delay time: actual train departure time (simulation log time) - expected train departure time (train timetable)

OC avg processing time (no delay): train departure time - 1st OC pick up time ($\approx$ 1st IC arrival time)

OC avg terminal time: OC avg delay time + OC avg processing time (no delay)

OC emissions: weighted vehicle lifts + trips

Table 2 Experimental Factors and Levels

	SPARCs	CTs
Train Batch Size	20, 30, 40, 50	100, 200
Number of Trains	1, 2, ..., 20	1, 2, ..., 10
Daily Throughput	40 to 2000	200 to 4000
Number of Cranes	1, 2, 3, 4	1, 2, 3, 4
Number of Hostlers	1, 2, 3, 4, ..., 8, 16, 32	1, 2, 3, 4, ..., 8, 16, 32

FINDINGS AND CONCLUSIONS

Table 3 Comparison of Required Terminal Size by Operation Type/Size (400 containers/day)

	SPARCs Lower bound	SPARCs Upper bound	CTs Lower bound	CTs Upper bound
Daily Throughput	400	400	400	400
Train Batch Size	20	50	100	200
Platoons/Trains per Day	10	4	2	1
Rows (M)	2	2	2	2
Columns (N)	2	2	3	3
Pairs of parking slots (n_r)	6	13	17	34
Yard Length (A; ft)	300	440	520	860
Yard Width (B; ft)	340	340	480	480
Footprint (A*B; ft ²)	102,000	149,600	249,600	412,800

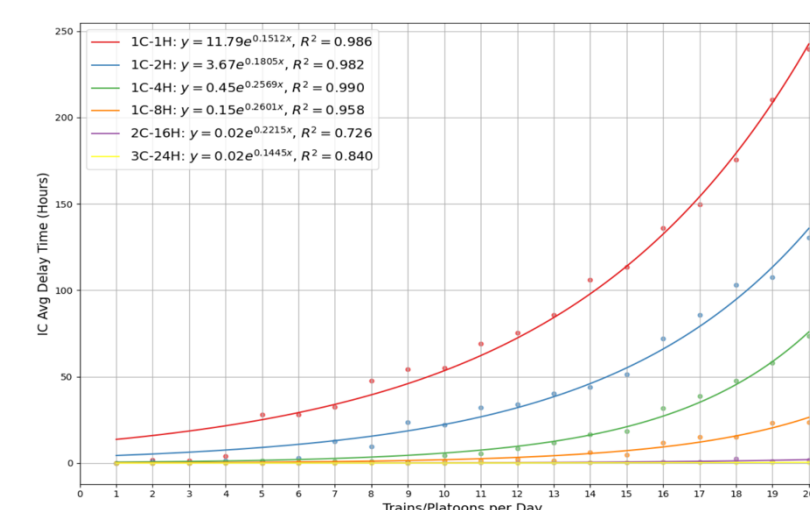


Fig. 9: Terminal delay (20-vehicle SPARCs)

Processing Time Delays

- Increasing trains/platoons per day leads to exponential increases in terminal time due to queuing delays
- In the congested stage, terminal performance is highly sensitive to train/platoon arrival patterns
- In the saturated state, the container arrival rate exceeds the processing rate as the cranes and hostlers become cannot reload the train/platoon fast enough
- Additional hostlers yield minimal benefit without adding cranes or additional terminal tracks

Table 4 Comparison of Terminal Processing Performance by Operation Type

Operation	Container Type	Processing Performance Function Coefficient (b_1)	R ²
SPARCs	IC	$0.001 - 0.005 \times \frac{1}{c} + 0.15 \times \frac{1}{H}$	0.98
	OC	$0.001 - 0.001 \times \frac{1}{c} + 0.15 \times \frac{1}{H}$	0.98
CTs	IC	$0.001 - 0.003 \times \frac{1}{c} + 0.15 \times \frac{1}{H}$	0.97
	OC	$0.005 - 0.003 \times \frac{1}{c} + 0.15 \times \frac{1}{H}$	0.99

Conclusions

Spatial Efficiency

- SPARCs require significantly less container terminal space than CTs at the same throughput.
- 200-car CT requires nearly **3 times** the optimal footprint of a 20-SPARC platoon.

Energy Efficiency

- At 400 containers/day $\times$ 365 days/year, SPARC terminal shows significant emissions reductions.
- SPARC terminal saves approximately **43,000 gallons** diesel annually.

Operational Performance

- SPARCs show **flatter delay function growth** and **lower sensitivity to resource shortages**.
- High-frequency, small-batch arrivals make SPARCs **more robust under limited resources**.

Resource Efficiency

- SPARC terminal requires **fewer resources**, further reducing capital cost of small footprint

LIMITATIONS AND FUTURE WORK

Study Limitations

- Assumes balanced inbound and outbound container flows
- Assumes uniform container demand and equal container priority
- Does not consider container stacking operations
- Train arrivals assumed random rather than schedule-optimized
- Terminal modeled as a generic representative facility

Future Work

- Evaluate unbalanced flows and time-varying container demand
- Optimize SPARC platoon sizing under non-uniform demand patterns
- Incorporate container stacking and priority handling effects
- Extend LIFTS to multi-track and multi-train terminal operations
- Integrate LIFTS with mainline simulations of SPARCs and CTs
- Compare end-to-end performance, energy, and emissions of SPARCs with diesel and battery trucks

This project is sponsored by the US Department of Energy: Advanced Research Projects Agency – Energy (ARPA-E), and technical collaboration and assistance from:

